Introduction to Robotics, Homework Hard 1

Manipulation: UR5, IIMT CI-05, Limo CoRoT arm

A pallet of cubes from a taught frame

MIPT

Center for Cognitive Modeling

What this homework is

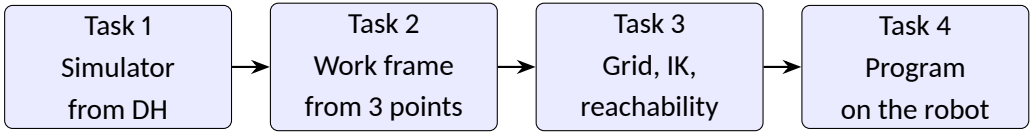
Make an arm place cubes on a grid it has never been shown

The arm builds a 3 × 3 pallet of cubes on a table. You do not read the nine cell positions off the controller one by one: you compute them in a coordinate frame that you teach the robot by touching three points, and you check every pose in a simulator of your own before the real arm moves.

- Choose **one** arm: UR5, IIMT CI-05, or the Limo CoRoT arm.
- Name the arm on the first page of your report.
- If you choose the Limo arm, use the Limo robot in Homework Hard 2 as well.

Four tasks, one chain

Each task produces the input of the next one



The simulator (Task 1) checks the grid (Task 3). The grid is written in the work frame (Task 2). The program (Task 4) runs the same grid on the real arm, and you measure how well the real arm repeats it.

UR5 (Universal Robots)



What it is. Six-joint collaborative arm, payload 5 kg, reach 850 mm; ours has a two-finger gripper.

Programming. On the teach pendant: Move blocks with waypoints, a Loop and Features (coordinate frames). The pendant shows the TCP pose.

Documentation. universal-robots.com/manuals; the UR page “DH parameters for calculations of kinematics and dynamics”.

IIMT CI-05



What it is. Six-joint collaborative arm built like the UR5 but with its own dimensions: payload 5 kg, working radius 900 mm, mass 20.4 kg, repeatability ±0.03 mm, joints ±360°. Two units, each with its own controller and pendant.

Programming. On its teach pendant, with waypoints, a loop and a user coordinate frame, as on the UR5.

Documentation. iimtcobot.com/iimt-ci-05-robotic-arm.html; on the downloads page of the same site: the CI-05 outline drawing.

Limo CoRoT arm (myCobot 280)



What it is. A small six-joint arm on the Limo base: reach 280 mm, payload 250 g, repeatability ±0.5 mm, adaptive gripper; use light cubes. Joint limits J1 ±168°, J2 ±135°, J3 ±150°, J4 ±145°, J5 ±165°, J6 ±180°.

Programming. No pendant: a Python script on the Limo computer sends joint angles (degrees) or poses (millimetres, degrees) through pymycobot; see limo_arm.py.

Documentation. github.com/elephantrobotics/LIMO-Cobot-docs, folder mycobot_280_M5(2023)en: parameters, Python API, DH figure.

Link dimensions and DH parameters

The numbers for Task 1, in millimetres

In the standard Denavit–Hartenberg convention joint i has a link length a_i (along the common normal of two joint axes), a link twist α_i, a joint offset d_i (along the joint axis) and a joint angle θ_i, the variable of a revolute joint.

	d_1	a_2	a_3	d_4	d_5	d_6
UR5	89.159	−425	−392.25	109.15	94.65	82.3
IIMT CI-05	139	−394.5	−381	138	143	103
Limo arm	131.56	110.4	96	64.62	73.18	48.6

UR5 and IIMT: $\alpha = (90°, 0, 0, 90°, -90°, 0)$; every other a_i and d_i is zero. The Limo arm is given by lengths only; its frames, signs and angle offsets are in the vendor figure 280DH.jpg of the documentation.

Task 1: Simulator from DH

Build the model of your arm

From the previous table, in any language:

- **FK:** a function from a joint vector q to the 4×4 pose of the *flange* (the end of the last link) in the base frame. The *TCP* (tool centre point: the point of the gripper that touches the cube) is the flange pose times a fixed tool transform that you measure. Set the same TCP on the controller.
- **Drawing:** the arm as a chain of joint positions for any q.
- **Limits:** a warning for a q outside the joint ranges.

dh_fk.py: UR5 reference positions to match to 0.1 mm; other arms, a tape measure.

Task 1: What to bring

- The three comparisons with the UR5 reference, or your own proof for another arm.
- Drawings of the arm in two poses.
- A photo of the real arm in one of those two poses.

Task 2: Work frame from three taught points

The arm must be told where the table is

A *work frame* is a coordinate frame glued to the table: an origin and two axes.

- Mark three points: the origin P_0, a point P_1 on the $+X$ axis, a point P_2 on the $+Y$ side.
- Bring the TCP to each mark and read its position (pendant screen, or `get_coords()` on the Limo arm).
- Compute the 4×4 matrix of the work frame in the base frame; show that its rotation part is a proper rotation.
- UR5 and IIMT: create the three-point frame on the pendant as well and compare.

Task 2: What to bring

- Photos of the TCP at the three marks.
- The three readings and your matrix.
- How far its rotation part is from a perfect rotation.
- UR5 and IIMT: the difference to the pendant’s own frame.

Task 3: Grid and reachability

Nine cells in the work frame

The spacing is yours; it must leave room for the gripper fingers.

- Each cell has three poses: *approach* (above), *grasp or place*, *retreat*. The gripper points down.
- Solve the inverse kinematics of every pose in your simulator; say which of the solutions you chose and why.
- Report for every pose the smallest singular value $\sigma_{\min}$ of the Jacobian (closeness to a singularity) and the distance of each joint to its limit.
- Replay the whole sequence in the simulator.

Task 3: What to bring

- A table of the 27 poses, with $\sigma_{\min}$ and the joint margins.
- A map of $\sigma_{\min}$ over the grid and a screenshot of the replay.
- What you changed when a pose was unreachable or near a singularity.
- Limo arm: the reach is only 280 mm, so choose the grid to fit.

Task 4: Robot program

Ten cycles, then measure

Write the pick-and-place program on the controller (Limo arm: a Python script with limo_arm.py) and run ten cycles.

- Straight-line moves for approach and retreat; joint or blended moves between cells; a loop for the cycles.
- Mark on paper where each cube stands after every cycle; measure the scatter with a ruler.
- Time one cycle with blend radius 0 and one with a blend radius above 0. Limo arm: it has no blend radius, so time a cycle that waits at every waypoint and a cycle that sends the next command at once.

Task 4: What to bring

- Screenshots of the program and its waypoints.
- Photos of the placed cubes and of the marked paper.
- The scatter, the two cycle times, and why they differ.

What you hand in

The same rules for all four tasks

One report. The first page names your arm and shows a photo of the setup. For each task:

- what you did and why, with the formulas and numbers;
- photos of the setup and the result, with a ruler or tape in frame wherever a distance is measured;
- screenshots of the program or code, printed values and plots;
- a table of several runs, and what went wrong and what you changed.

Code goes as files; the report quotes only what it discusses.

Deserve “A” grade!

– Oleg Bulichev

bulichev.ov@mipt.ru

@Lupasic

Цифра, 521